

Math 241, F12  
Quiz 2

Name: Answer  
SSN: xxx-xx-

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz.

**Problem 1.** (10 points) Given two vectors  $\mathbf{a} = \langle 3, 6, -2 \rangle$  and  $\mathbf{b} = \langle 4, 0, 3 \rangle$ .

(a) Find the angle between  $\mathbf{a}$  and  $\mathbf{b}$ .

$$\begin{aligned} |\mathbf{a}| &= \sqrt{3^2 + 6^2 + (-2)^2} = \sqrt{49} = 7 & \mathbf{a} \cdot \mathbf{b} &= 3 \cdot 4 + 6 \cdot 0 + (-2) \cdot 3 = 6 \\ |\mathbf{b}| &= \sqrt{4^2 + 0^2 + 3^2} = \sqrt{25} = 5 \\ \cos \theta &= \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| |\mathbf{b}|} = \frac{6}{7 \cdot 5} = \frac{6}{35} \\ \Rightarrow \theta &= \cos^{-1}\left(\frac{6}{35}\right) \end{aligned}$$

(b) Find the vector projection of  $\mathbf{b}$  onto  $\mathbf{a}$ ,  $\text{proj}_{\mathbf{a}} \mathbf{b}$ .

$$\begin{aligned} \text{proj}_{\mathbf{a}} \mathbf{b} &= \left( \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}|^2} \right) \mathbf{a} \\ &= \left( \frac{6}{7^2} \right) \langle 3, 6, -2 \rangle \\ &= \left\langle \frac{18}{49}, \frac{36}{49}, -\frac{12}{49} \right\rangle \end{aligned}$$

(c) Compute the cross product  $\mathbf{a} \times \mathbf{b}$ .

$$\begin{aligned} \mathbf{a} \times \mathbf{b} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 6 & -2 \\ 4 & 0 & 3 \end{vmatrix} = \begin{vmatrix} 6 & -2 \\ 0 & 3 \end{vmatrix} \hat{i} - \begin{vmatrix} 3 & -2 \\ 4 & 3 \end{vmatrix} \hat{j} + \begin{vmatrix} 3 & 6 \\ 4 & 0 \end{vmatrix} \hat{k} \\ &= 18\hat{i} - (9 - (-8))\hat{j} + (-24)\hat{k} \\ &= 18\hat{i} - 17\hat{j} - 24\hat{k} \end{aligned}$$